

BY 600: Research Methodology

Course Code	BY 600
Course Name	Research Methodology
L-T-P-C	1-0-0-1
Prerequisites	NA
Intended for	Masters and PhD candidates
Mutual Exclusion	NA
Approval	53rd BoA (Dec 2023); 42nd Senate (Jan 2024), in the M.Tech Biotechnology curriculum

Course Contents

- What is research The concept and objectives of research, types of research, research hypothesis.
- Research planning or experimental design Necessity to define research problem, research gap, working hypothesis, planning process, research design.
- Lab and biosafety Safe laboratory practices, biosafety levels, biological hazard information, labels, signs and storage of chemical and biological reagents, Classes and types of biosafety cabinets, inactivation and disposal of biological products, waste management, and incident response, report and risk management.
- Data collection and analysis Basics of statistics, data processing and analysis strategies and tools, hypothesis testing, interpretation of data.
- Literature search, collection, storage and citation Importance of literature review in defining a problem, literature review, online tools.
- Writing research reports Types of reports (articles, reviews, progress reports and thesis), formats, planning of report writing. Figure preparation and manipulation. Effective use of English language in scientific papers.
- Science journals and the peer-review process Types of journals, publication process, peer-review process, meaning of impact factor, Hindex, etc.
- Presentation skills Preparation of slides for talks, preparation of posters. Presentation types and skills.
- Research ethics Ethical issues, citation and acknowledgement, plagiarism, reproducibility and accountability.
- IPR Intellectual property rights and patent law, commercialization, copy right, trademarks, royalty, trade related aspects of intellectual property rights and a few case studies.
- Science popularization and outreach Media coverage, lectures in public forums, circulation of scientific ideas and provoking thoughts, and the pros abs cons of these activities.

References

1. Wyne C. Booth, Colomb, William, **The Craft of Research**, 4th Edition, University of Chicago Press.
2. Judith Bell, Stephen Waters, **Doing Your Research Project: A Guide for First-time Researchers**, McGraw-Hill Education, 2014.
3. Ranjit Kumar, **Research Methodology: A Step-By-Step Guide For Beginners**, Sage South Asia, 2011.
4. C. R Kothari, **Research Methodology: Methods and Techniques**, New Age International, 2004.
5. Laura Hyatt, Carol Roberts, **The Dissertation Journey**, Corwin, 2023.
6. Jari Saramaki, **How to Write a Scientific Paper: An Academic Self-Help Guide for PhD Students**, 2018.
7. WIPO Intellectual Property Handbook (Web resources).
8. The WHO Laboratory Biosafety Manual (LBM), 4th edition (online).